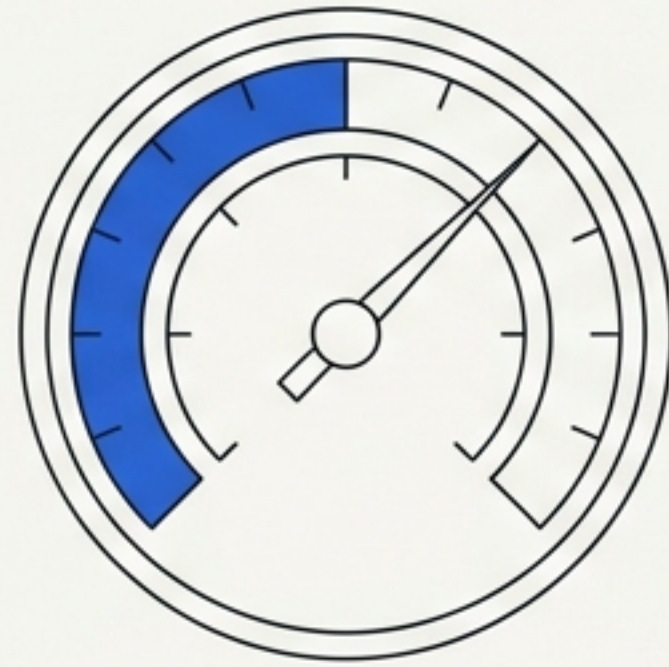


A system that runs is not a system that works.



Phase 5 / Measuring RAG Success through Telemetry and Feedback.
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Without telemetry, engineering decisions are just guesses.

Without Evaluation

[- Cannot tell if a code change improved or degraded performance.]

[- Cannot identify specific topics failing in production.]

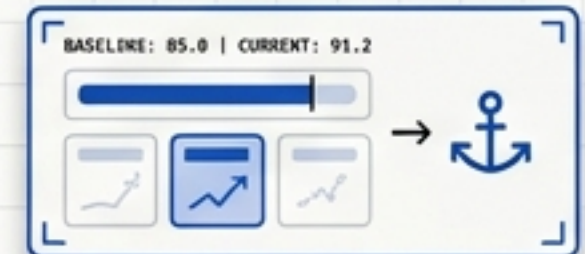
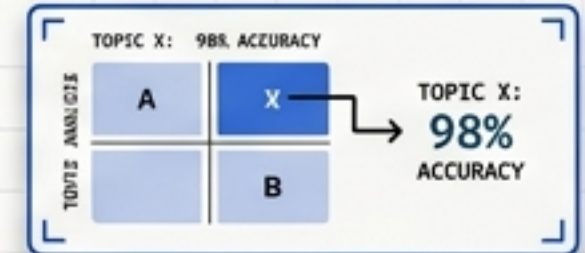
[- Cannot report system quality to stakeholders.]

With Evaluation

[✓ Every improvement becomes a measurable metric.]

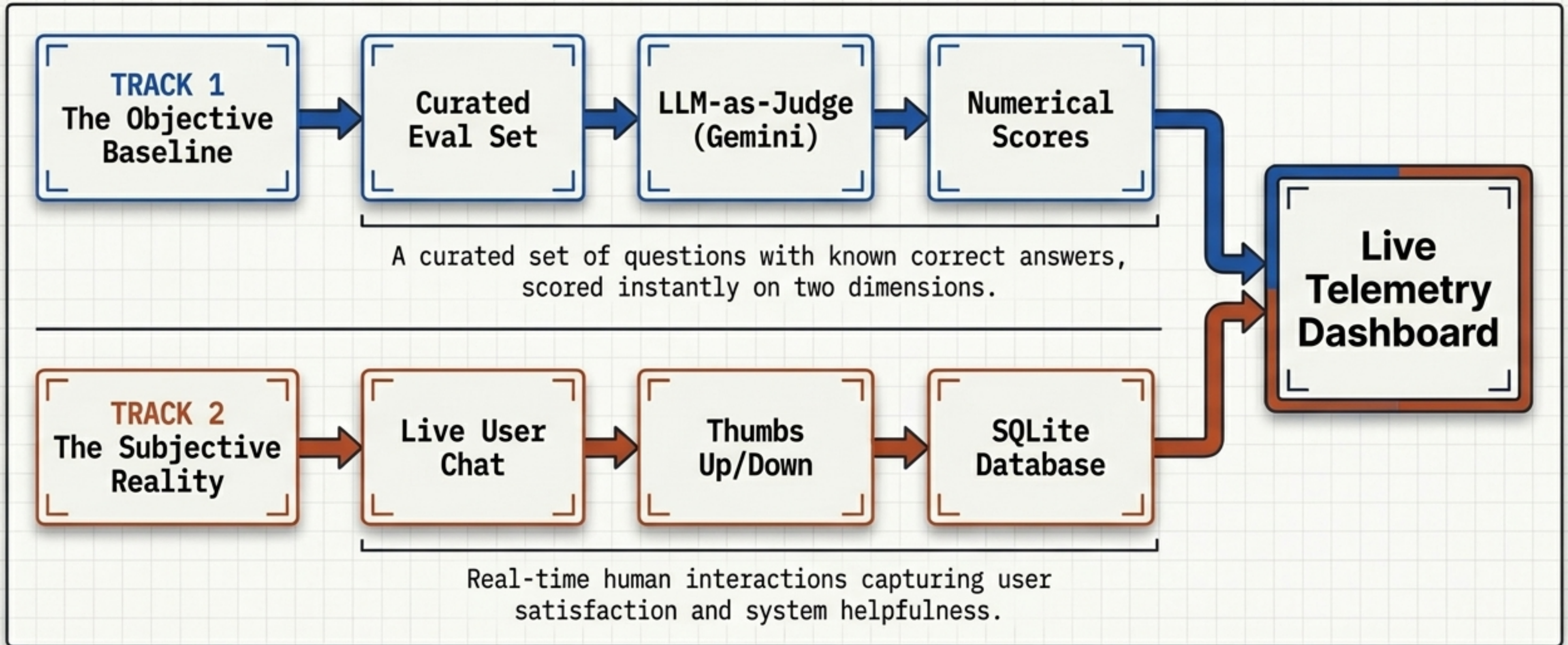
[✓ Weak areas are isolated by specific topic.]

[✓ The baseline score anchors every future architectural change.]



In production AI systems, **evaluation is not optional**. It is the framework for all engineering decisions.

The Dual-Track Telemetry Architecture



The foundation of automated telemetry is a curated ground truth.

EVAL_SET_V1

20 hand-written questions extracted from actual Holberton documents.

[Academic Standing]

[Dismissal]

[Campus Transfers]

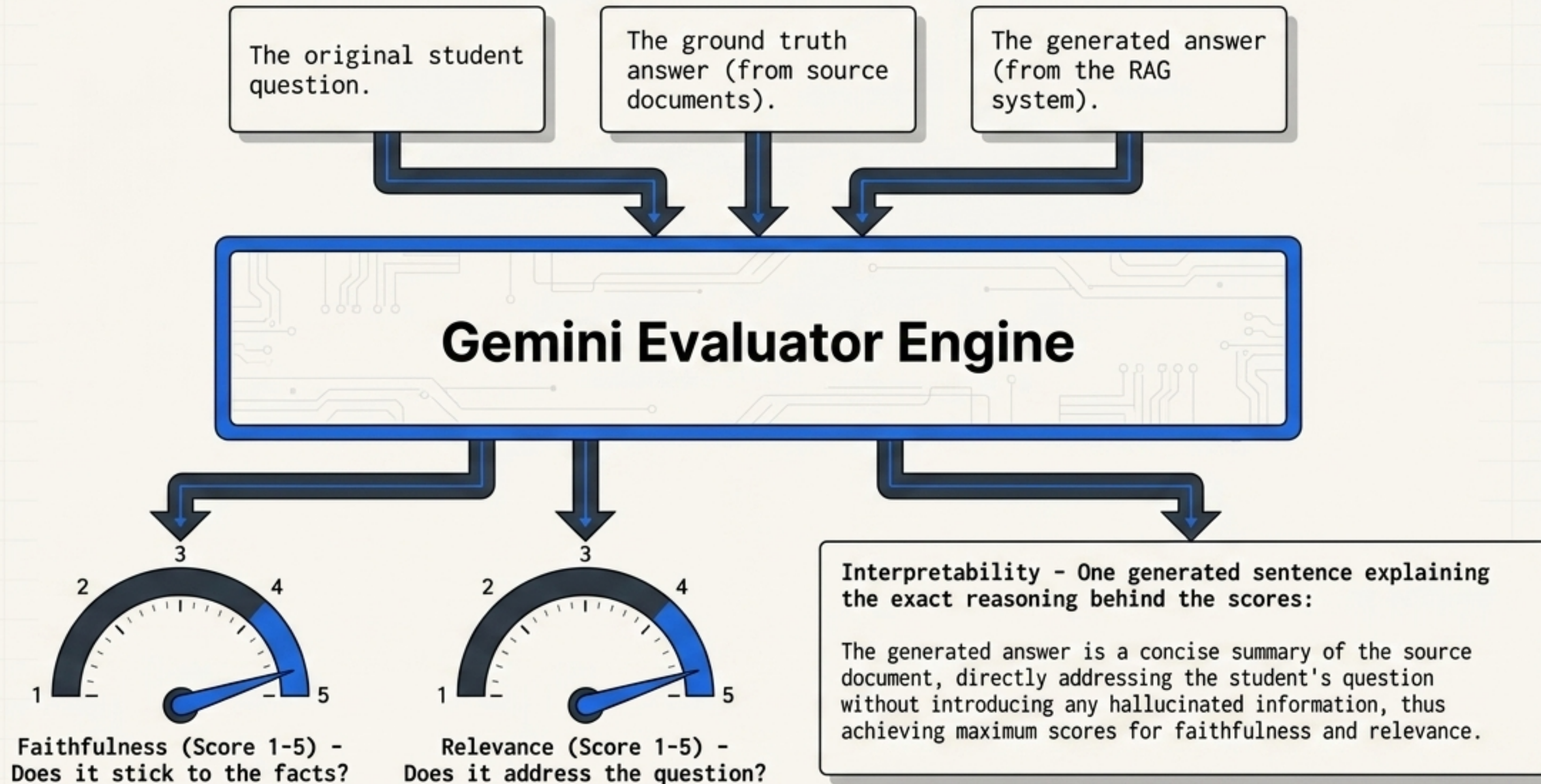
[Project Scoring]

[Mock Interviews]

Why hand-written?

Auto-generated questions produce auto-generated answers that score well against each other and mean nothing. A hand-written question with a known correct answer is the only honest measurement of the system's retrieval capabilities.

The LLM-as-Judge processing funnel



Diagnosing the two primary failure modes

Relevance
(Retrieval)

"True but useless." (Retrieval Failure)

Asked: "How do I contact my campus director?"
Answer: "The minimum average is 80%."
(Factually true, entirely unhelpful).

"On-topic but hallucinated." (Grounding Failure)

Asked: "What are the transfer requirements?"
Answer: "You can transfer anytime with a 70% average."
(Addresses the topic, but the facts are invented/wrong).

Faithfulness (Grounding)

High faithfulness confirms grounding. **High relevance** confirms retrieval. A production RAG system requires both.

Establishing the Phase 5 baseline metrics

4.78 / 5

Overall Score:

4.70 / 5

Avg Faithfulness:

4.85 / 5

Avg Relevance:

Perfect Scores (5.0)

Academic standing

dismissal

cheating

mock interviews

peer learning

specializations

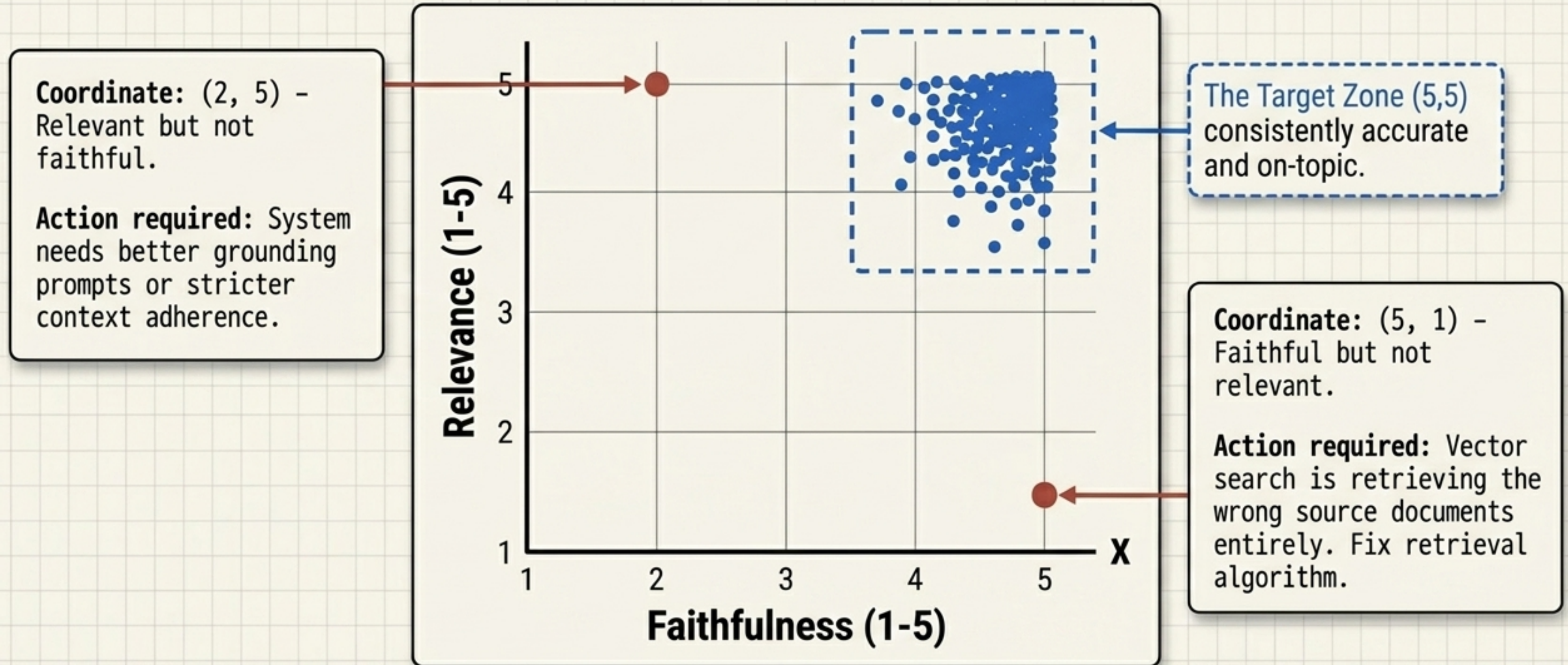
The Phase 6 Roadmap (Needs Work)

Campus transfers (4.5): Short document, retrieval imprecise.

Project scoring (4.0): Spans multiple docs, synthesis required.

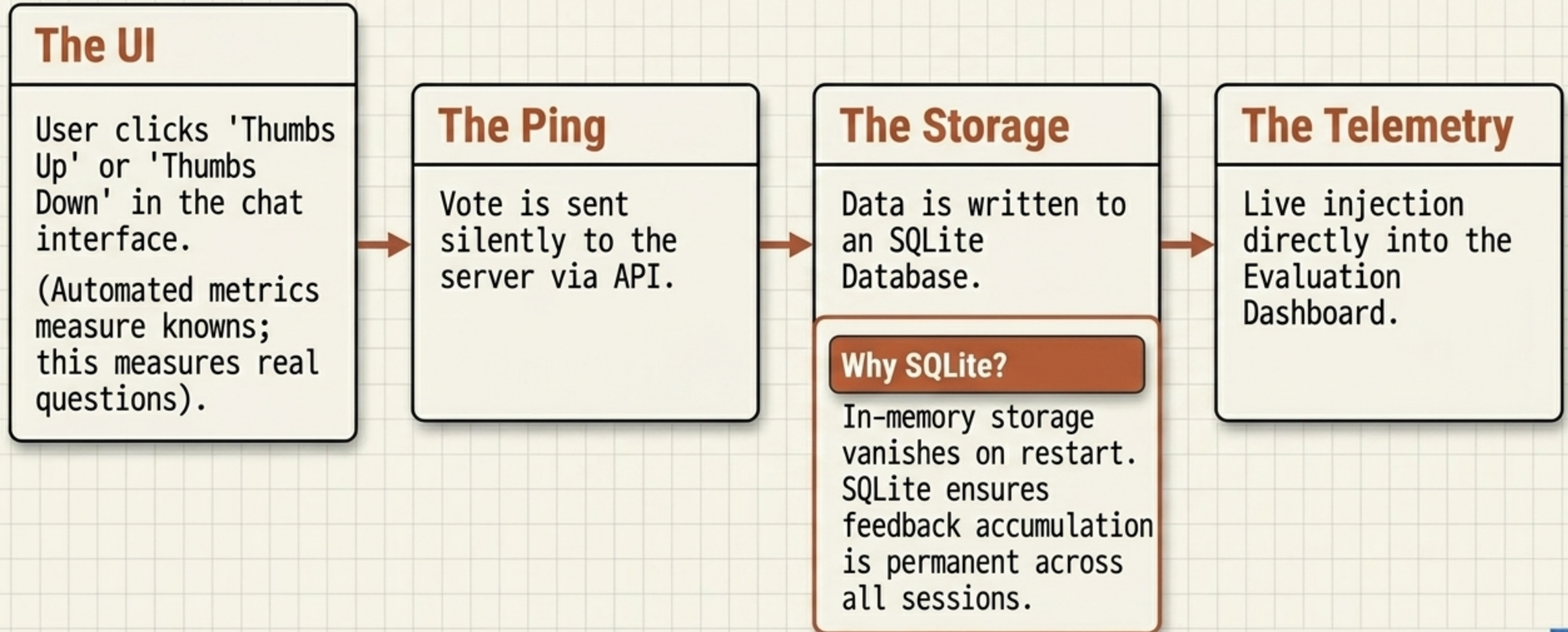
Evaluation quizzes (4.0): Specific numbers sometimes paraphrased.

The Telemetry Scatter Plot: Outliers define the work



One outlier point tells an engineer more than the overall average score.

Track 2: Capturing the subjective reality of users



Two distinct signals measuring different realities

Automated Evaluation		Human Feedback	
Scope:	Known questions and expected behaviors.	Scope:	Unanticipated, diverse, and real student queries.
Speed:	Fast (20 questions processed in minutes).	Speed:	Slow (Accumulates over days and weeks).
Nature:	Reproducible, comparable across versions, but blind to user intent.	Nature:	Subjective, messy, but reflects actual utility.
Goal:	Measures Objective Accuracy.	Goal:	Measures User Satisfaction.

The Synthesis: Why production AI requires both signals



**Automated evaluation measures what you built.
Human feedback measures what users actually need.**

A system that scores a perfect 5.0 on automated metrics but receives a wave of human “thumbs down” is not a success. It is a system that successfully solved the wrong problem. The eval set cannot see this failure. Only the dual-track framework surfaces the truth.

Telemetry principles for every AI deployment

1. Measure Before You Improve

Without a baseline, every change is a guess. The **4.78** score isn't just a metric; it is the mathematical reference point for validating all **Phase 6** optimizations.

2. Evaluation is a Design Decision

The eval set chosen, the dimensions measured, and the judge model utilized shape the very definition of 'quality'. Different choices produce different numbers.

3. Dual Signals are Mandatory

No eval set anticipates every question. No human feedback is reproducible enough for regression testing.

Production systems run both tracks, always.

